Light

GCSE Computer Science - Character Sets (Set 3)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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← Back to Character Sets

Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What is a character code?

- **A.** A secret password for accessing files
- **B.** A unique numeric value assigned to represent a character
- **C.** A type of programming language
- **D.** A method for compressing images

Q2 (1 mark)

When you save a text document, what is actually stored on the storage device?

- **A.** Photographs of each letter

- B. Sound recordings of each word
- C. The font styles and colours only
- D. The binary codes representing each character

Q3 (1 mark)

ASCII was primarily designed to represent text in which language?

- A. English
- B. Chinese
- C. Arabic
- D. Japanese

Q4 (1 mark)

In ASCII, the letter 'Z' has the code 90. What is the code for 'Y'?

- A. 91
- B. 89
- C. 88
- D. 92

Q5 (1 mark)

What does 'encoding' mean in the context of character sets?

- ○ **A.** Making text invisible
- ○ **B.** Deleting unwanted characters
- ○ **C.** Converting characters into their numeric code values
- ○ **D.** Changing the font of text

Q6 (1 mark)

In Unicode, each character is assigned a unique identifier called a:

- ○ **A.** Pixel
- ○ **B.** Byte
- ○ **C.** File name
- ○ **D.** Code point

Q7 (1 mark)

Why can ASCII not represent characters like é, ñ, or ü?

- ○ **A.** These characters are too complex to display
- ○ **B.** These characters require colour displays
- ○ **C.** ASCII only includes basic Latin letters without accents or diacritics
- ○ **D.** ASCII files cannot be saved on computers

Q8 (1 mark)

In ASCII, printable characters (visible symbols) begin at code 32. What character has code 32?

- ○ **A.** The space character
- ○ **B.** The letter 'A'
- ○ **C.** The digit '0'
- ○ **D.** The exclamation mark '!'

Q9 (1 mark)

Although 7-bit ASCII only needs 7 bits per character, how many bits are typically used when storing each ASCII character in a file?

- ○ **A.** 4 bits
- ○ **B.** 7 bits exactly
- ○ **C.** 16 bits
- ○ **D.** 8 bits (1 byte)

Q10 (1 mark)

A student wants to include the copyright symbol © in their document. Which encoding standard includes this symbol?

- ○ **A.** Unicode
- ○ **B.** 7-bit ASCII only
- ○ **C.** Neither ASCII nor Unicode

- ○ **D.** Only image formats, not character encoding

Q11 (1 mark)

In ASCII, 'z' (lowercase) has the code 122. What is the code for 'x'?

- ○ **A.** 118
- ○ **B.** 119
- ○ **C.** 121
- ○ **D.** 120

Q12 (1 mark)

Why is it important that character sets are standardised internationally?

- ○ **A.** To make computers run faster
- ○ **B.** To reduce the cost of keyboards
- ○ **C.** To ensure text appears correctly when shared between different systems
- ○ **D.** To limit the number of available characters

Q13 (2 marks)

In ASCII, 'A' = 65. Calculate the code for the letter 'L'.

- ○ **A.** 75

- B. 77
- C. 76
- D. 78

Q14 (2 marks)

A character has ASCII code 116. Given that 'a' = 97, which lowercase letter does code 116 represent?

- A. t
- B. r
- C. s
- D. u

Q15 (2 marks)

How many bits are required to store the word 'SUN' using 7-bit ASCII?

- A. 14 bits
- B. 24 bits
- C. 21 bits
- D. 28 bits

Q16 (2 marks)

The binary value 01001000 represents an ASCII character. Given that 'A' = 65, which character is this?

- ○ **A.** E
- ○ **B.** F
- ○ **C.** G
- ○ **D.** H

Q17 (2 marks)

A user opens a document and sees □□□ or ??? instead of the expected text. What is the most likely cause?

- ○ **A.** The system does not have the character encoding or font needed to display those characters
- ○ **B.** The computer's processor is too slow
- ○ **C.** The document is password protected
- ○ **D.** The file is too large to open

Q18 (2 marks)

In ASCII, 'E' = 69. Using the fact that lowercase letters are 32 higher than uppercase, what is the code for 'e'?

- ○ **A.** 37
- ○ **B.** 91
- ○ **C.** 99

- D. 101

Q19 (2 marks)

The ASCII code for 'Q' is 81. What is 81 in binary?

- A. 0100 1111
- B. 0101 0011
- C. 0101 0001
- D. 0101 1000

Q20 (2 marks)

Given that '0' = 48 in ASCII, what is the ASCII code for the digit character '9'?

- A. 9
- B. 57
- C. 55
- D. 58

Q21 (2 marks)

A message contains exactly 100 characters. What is the difference in storage size between 7-bit ASCII and 16-bit Unicode for this message?

- A. 700 bits

- B. 1600 bits
- C. 900 bits
- D. 2300 bits

Q22 (2 marks)

ASCII code 100 represents a lowercase letter. Given 'a' = 97, which letter is it?

- A. c
- B. e
- C. d
- D. f

Q23 (2 marks)

When you send an email, why does the email system need to use character encoding?

- A. To make emails arrive faster
- B. To add colours to the email text
- C. To convert text characters into binary data that can be transmitted over networks
- D. To reduce the cost of sending emails

Q24 (3 marks)

Calculate the sum of ASCII codes for the word 'FAN'. Given that 'A' = 65.

- ○ **A.** 208
- ○ **B.** 213
- ○ **C.** 211
- ○ **D.** 218

Q25 (3 marks)

A novel contains 500,000 characters. Calculate the storage size in kilobytes (KB) for both 7-bit ASCII and 16-bit Unicode. (Use 1 KB = 1000 bytes)

- ○ **A.** ASCII: 437.5 KB, Unicode: 1000 KB
- ○ **B.** ASCII: 500 KB, Unicode: 500 KB
- ○ **C.** ASCII: 3500 KB, Unicode: 8000 KB
- ○ **D.** ASCII: 62.5 KB, Unicode: 125 KB

Q26 (3 marks)

A Japanese user tries to enter their name in a program that only supports ASCII. Explain what problem they will face and why.

- ○ **A.** The name will display correctly but save incorrectly
- ○ **B.** The name will appear in a different font style
- ○ **C.** The name will be automatically translated to English

○ **D.** The program cannot accept or store Japanese characters because ASCII only includes 128 characters limited to the Latin alphabet, digits, and basic symbols - none of which cover Japanese kanji, hiragana, or katakana

Q27 (3 marks)

A student claims that the ASCII code for the digit character '7' is 55. Verify whether this is correct by calculating from '0' = 48.

○ **A.** Incorrect - the code should be 7

○ **B.** Correct - '7' = 48 + 7 = 55

○ **C.** Incorrect - the code should be 49

○ **D.** Incorrect - the code should be 103

Q28 (3 marks)

When Unicode was developed, designers made the first 128 code points identical to ASCII. Analyse why this decision was made, considering existing systems and documents.

○ **A.** It was a random coincidence with no planned purpose

○ **B.** Unicode needed exactly 128 characters and simply copied ASCII

○ **C.** ASCII was more advanced than Unicode for the first 128 characters

○ **D.** This backwards compatibility ensured that billions of existing ASCII documents and systems could work with Unicode without any modification, easing the transition to the new standard

Q29 (3 marks)

A company is developing a global chat application for users in India, China, France, and Brazil. Analyse the character encoding requirements and potential problems.

- ○ **A.** Unicode is essential because it includes Hindi (Devanagari), Chinese characters, French accented letters (é, è, ç), and Portuguese accented letters (ã, ç), enabling all users to communicate in their native scripts within one application
- ○ **B.** ASCII is sufficient since all countries use the same alphabet
- ○ **C.** Four separate applications are needed, one for each country's encoding
- ○ **D.** Users should only communicate in English to avoid encoding issues

Q30 (3 marks)

Using the fact that 'A' = 65 and 'a' = 97, calculate the ASCII codes for 'D', 'd', and determine their sum.

- ○ **A.** D = 67, d = 99, sum = 166
- ○ **B.** D = 68, d = 100, sum = 168
- ○ **C.** D = 69, d = 101, sum = 170
- ○ **D.** D = 68, d = 132, sum = 200

Total: **55** marks • Time: **30 min**

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